Brad Stricherz

Senior Geospatial Software Engineer

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St. Louis, MO | Active TS/SCI Clearance

Professional Summary

Senior software engineer with 15+ years in geospatial intelligence and WebGIS development, pairing a GEOINT analyst's command of spatial data with full-stack engineering in React, TypeScript, Python, and PostGIS. At Gateway Geospatial Group (G3), builds a containerized geospatial data platform for a defense customer, deployed to OpenShift on classified networks. Spent 15 years at the National Geospatial-Intelligence Agency, progressing from imagery analyst to full-stack developer on systems used in U.S. Government operations worldwide. Founder of GeoBrad.dev, an open-source geospatial practice serving nonprofits and researchers. M.S. in Spatial Data Science, Penn State. Active TS/SCI clearance.

Technical Skills

Geospatial: PostGIS, GeoDjango, OGC services, ArcGIS REST, STAC, GeoPandas, ArcPy, PyQGIS, GEOINT, Imagery Analysis, Spatial Analysis

Mapping: OpenLayers, Leaflet, CesiumJS

Frontend: TypeScript, JavaScript, React, React Native, Chakra UI

Backend and Data: Python, Django, Django Ninja, FastAPI, GraphQL, Node.js, Apache Airflow

Databases and Search: PostgreSQL, Trino, Neo4j, OpenSearch, SQL

DevOps: Docker, OpenShift, Helm, GitLab CI/CD, GitHub Actions, Git

Professional Experience

Senior Software Engineer, Gateway Geospatial Group (G3) — Apr 2026 - Present

Build full-stack features for a containerized geospatial data platform serving a defense customer, across React/TypeScript, GraphQL, and Python/Airflow.

Developed a self-service onboarding wizard that registers OGC, ArcGIS REST, STAC, and database sources, with secure-by-design credential handling and SSRF protection.

Write ingestion pipelines into a federated Trino, Neo4j, and OpenSearch stack, and built an operations console for pipeline monitoring.

Led the admin UI migration to Chakra UI v3, improving accessibility and performance, and enforce a 95% test coverage gate across three applications.

Support GitLab CI/CD pipelines deploying via Helm to OpenShift on classified networks, and remediate security scan findings for accreditation.

Established an AI-assisted, multi-agent development workflow.

Founder and Geospatial Developer, GeoBrad.dev — 2024 - Present

Founded an open-source geospatial practice delivering mapping and spatial analysis tools to nonprofits, researchers, and community organizers, mostly pro bono and built so each organization can run it without a GIS team.

Built and open-sourced a full-stack WebGIS template suite: a Django Ninja/PostGIS backend, a React/Leaflet web frontend, and a React Native/Expo mobile app that runs against it.

Built QuakeGlobe, a real-time 3D earthquake visualizer using CesiumJS, React, and the USGS Earthquake API, with magnitude and depth filters and animated playback.

Automated extraction and geospatial matching of St. Louis property tax auction data, parsing sheriff sale PDFs and fuzzy-matching addresses to parcels with GeoPandas.

Full-Stack Software Developer | Geospatial and Imagery Analyst, National Geospatial-Intelligence Agency **2010 - Sep 2025**

Progressed from GEOINT and imagery analyst to full-stack developer over 15 years, building and supporting systems used in U.S. Government operations worldwide.

Developed web applications for the Aeronautical Safety and Navigation Office using React, FastAPI, and PostGIS.

Built interactive mapping tools and microservices for aeronautical safety visualization and workflow automation.

Shipped applications through GitLab CI/CD and Docker on PostgreSQL/PostGIS backends.

Produced geospatial intelligence for time-sensitive operations and targeting missions supporting the FBI and DIA.

Deployed as a GEOINT analyst to embassies and military commands in Iraq, Saudi Arabia, and South Korea, and aboard the U.S. Navy's 7th Fleet.

Education

Master of Science in Spatial Data Science, Penn State University, 2023

Master of Arts in International Relations, Webster University, 2019

Bachelor of Science in Geography, South Dakota State University, 2009

Certifications

Cesium Certified Developer

DoD Certified Python Instructor

Penn State: Geospatial Programming and Web Map Development

GEOINT Professional Certifications

Docker Foundations Professional